7.

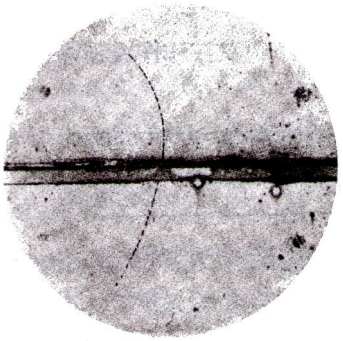
1. In the Bohr model of ground-state helium, each electron circles the nucleus, with equal radii and angular momenta $\hbar$: $\hbar = h/2\pi$, where h is the Planck constant.

(I) Ignore electron interaction. Find first ionization energy (eV): energy needed to move one electron to infinity.

(II) The measured ionization energy is 24.6eV . Include electron interaction in this model. Assume the two electrons always occupy opposite ends of a diameter through the nucleus. Find orbit radius r_0, ground energy E_0, ionization energy E^+. Compare the predicted first ionization energy with experiment.

The mass is m =0 .5MeV c^2, c is light speed; $\hbar c$ =197.3MeV7 .3eV • nm, ke^2 =1 .4MeV .4eV nm , k is Coulomb's constant and e is unit charge.

2. The photo is a cloud-chamber track, in the paper plane. In its middle a lead plate is normal to paper. A steady uniform field goes into it. The particle charge magnitude is e : e =1 .60 $\times 10^{-19}$ C. The track radius below is r_d =2 0mm , and above is r_u =7 6 .0mm . In lead, the track makes θ =1 5 .0° to the normal; the lead is d =6 .00mm thick; use B =1 .00 and m =9 .1×10 $^{-31}$ kg 0 .511 MeV / c^2. Ignore gas drag on the particle.



(I) State the direction of motion and sign of charge.

(II) Find the average force in N on the plate while the particle passes through.

(III) Instead, an accelerator pulse of flux j =5 .00×10 ^{18}/s hits the plate. One pulse lasts τ =2.5 0s . Find the average force in N on the plate while the pulse passes through, and the heat in J absorbed by the plate during it.